

EXPERT SELECTION IN TRAFFIC DETECTION: A CONTROLLED STUDY OF DEVELOPMENT SUPPORT

Haowen Liu Lei Zhang

Capital Normal University

ABSTRACT

Combining traffic experts requires deciding when a second prediction should replace the first. We separate arbiter fitting, threshold selection and evaluation across application/family groups on 40,000 USTC-TFC2016 segments. Across 16 threshold pairs, we trace candidate distinguishability to outer-fold false alarms and misses. In one fold, Logistic’s unique numerical development optimum adds 77 false alarms and removes five misses relative to Logistic-0.5. Logistic-0.5’s positive point estimate against a same-trigger rule disappears after selection; directional thresholds show no consistent advantage over simple-rule or global-threshold controls. At matched quotas, high- and low-coverage subsets select different policies in one fold: high coverage has fewer histogram-based gradient boosting (HGB) arbiter errors but more Logistic errors. Replacing Stats with Extra-Trees yields both transfer loss without tied-candidate divergence and divergence without loss, relative to each arbiter at 0.5. These cases separate candidate distinguishability from outer-fold error changes on this benchmark.

Index Terms— Traffic detection, expert selection, out-of-fold prediction, development support

1. INTRODUCTION

Expert complementarity does not ensure reliable policy selection. We relate candidate-level development information to outer-fold false alarms and misses. Identifier-filtered views may retain capture-specific timing/order cues [1].

Cascades and acquisition/selection balance accuracy and cost [2, 3]. META-DES learns competence from meta-features and low-consensus samples [4]; ECHO reports overturned early exits [5]. We fix experts and windows, unlike representation search [6].

Selection criteria can overfit [7]; underspecification and predictive multiplicity expose differences concealed by aggregate scores [8, 9]. We examine these risks in a fixed grid of traffic-selection policies under application/family shift.

We connect fitting support and candidate-vector equivalence to outer-fold errors. Matched quotas expose opposite arbiter responses; recall-filter removal separates changed feasibility from unchanged selection. Code and results are available [10].

2. TASK AND CONTROLLED PROTOCOL

2.1. Inputs and fitting roles

USTC-TFC2016 [11] supplies 2,000 stable-hash sampled segments per application/malware-family group (20 groups, 24 captures). Temporal uses up to 64 initial packets: 20 length/direction/timing features plus five per 2/4/8/16-packet prefix, without padding. Stats

uses eight completed-segment summaries: packet/byte counts, directional bytes/ratio, length mean/variance and duration. Both histogram-based gradient boosting (HGB) experts use 220 iterations, learning rate 0.05, 31 leaves, $L_2 = 0.1$, no early stopping, seed 42. Projection excludes labels, endpoints and provenance; labels supervise, provenance groups samples. This tests projection, not runtime FieldAudit or matched-prefix inputs (Fig. 1).

Five outer folds allocate 12/2/2/4 disjoint groups to fitting, reliability, selection and evaluation (24k/4k/4k/8k segments), class-balanced. Three inner group folds supply out-of-fold (OOF) disagreements for arbitration; prediction groups never fit experts/reliabilities. Outer samples are exposed. Stats ExtraTrees [12] (100 trees, Gini, square-root feature sampling, unrestricted depth, leaf size 1, no bootstrap, seed 42) regenerates OOF outputs, reliabilities and targets and refits both arbiters; Temporal stays frozen.

2.2. Fixed and learned decisions

Let p_T, p_S be malicious probabilities and $\hat{y}_j = \mathbf{1}[p_j \geq 0.5]$. Stats is read when $|p_T - 0.5| < 0.495$; otherwise, or on agreement, Temporal is retained. On triggered disagreement, Fixed switches if

$$|p_S - 0.5|r_S > 0.25 |p_T - 0.5|r_T. \quad (1)$$

Each inner/full expert’s reliability r_j is binary Macro-F1 at 0.5 on the shared independent 4k reliability role, not a calibrated probability. Constants 0.495 and 0.25 are inherited.

The arbiter fits $\mathbf{1}[\hat{y}_S = y]$ with uncalibrated output $q(z)$ from 11 inputs: both probabilities, margins, reliabilities and decisions; signed/absolute probability differences; and $p_T p_S$. Logistic regression uses training-only standardization, $C = 1$, at most 2,000 iterations. HGB uses 200 iterations, 15 leaves, minimum leaf size 30, $L_2 = 2$, native missing-feature handling, no early stopping. Both use seed 42.

HGB-0.5/Logistic-0.5 switch on $q \geq 0.5$. Dev16 selects $(\theta_0, \theta_1) \in \{0.25, 0.5, 0.75, 1.01\}^2$, indexed by $\hat{y}_T$; Global4 independently selects among the four diagonal pairs. Fixed-Dev16 replaces 0.25 in (1) with directional coefficients from $\{0.25, 0.5, 1, \infty\}^2$, using strict comparison; ∞ disables switching, avoiding $\infty \cdot 0$. Equal budgets need not imply equal statistical complexity.

Searches require recall at least Temporal’s (tolerance 10^{-12}), then maximize Macro-F1, recall and finally minimize switches; ties take the first ascending pair. No-switch is feasible; infeasibility raises an error. Controls, replay and backend analysis are exploratory with fixed roles/grids. All fold fits and 25 choices preceded ExtraTrees outer scoring.

2.3. Controls, metrics and uncertainty

Dev-expert (Temporal/Stats) and Dev-weighted (averaging weight) maximize selection Macro-F1 without the recall constraint. Weights

Policy	Stats HGB (original)					Stats ExtraTrees (sensitivity)				
	F1 (%)	Recall (%)	FPR (%)	C/D	Chg.	F1 (%)	Recall (%)	FPR (%)	C/D	Chg.
Temporal	92.833	98.720	13.005	0/0	0	92.833	98.720	13.005	0/0	0
Stats	86.003	96.915	24.585	208/2885	3093	79.903	99.220	38.020	971/5874	6845
Equal average	92.670	98.570	13.180	128/193	321	—	—	—	—	—
Dev-weighted	92.622	98.850	13.550	51/134	185	—	—	—	—	—
Dev-expert	87.238	98.295	23.515	59/2246	2305	—	—	—	—	—
Fixed	92.897	98.590	12.750	131/106	237	93.406	98.805	11.955	263/36	299
HGB-0.5	92.755	98.705	13.145	40/71	111	93.154	98.715	12.365	161/34	195
Logistic-0.5	93.007	98.805	12.745	107/38	145	92.962	98.710	12.740	83/32	115
HGB-Dev16	92.842	98.865	13.130	44/40	84	92.954	98.715	12.760	82/34	116
Logistic-Dev16	92.824	98.830	13.130	30/33	63	92.762	98.715	13.140	3/31	34
Fixed-Dev16	92.834	98.860	13.140	53/52	105	93.406	98.810	11.960	262/35	297
Logistic-Global4	92.832	98.830	13.115	30/30	60	92.762	98.715	13.140	3/31	34
HGB-Global4	92.849	98.865	13.115	44/37	81	92.957	98.710	12.750	81/32	113

Table 1. Same 40,000 segments and roles, separate Stats backends; Temporal unchanged. ExtraTrees uses fresh reliabilities/arbiters; —: not run. F1: binary Macro-F1. C/D: corrections/introduced errors versus Temporal. Chg.: prediction changes relative to Temporal (C + D). Calls: 1 for single-expert policies (including Dev-expert), 2 for averages, 1.104 for fixed/arbiter policies; policy reads, not runtime.

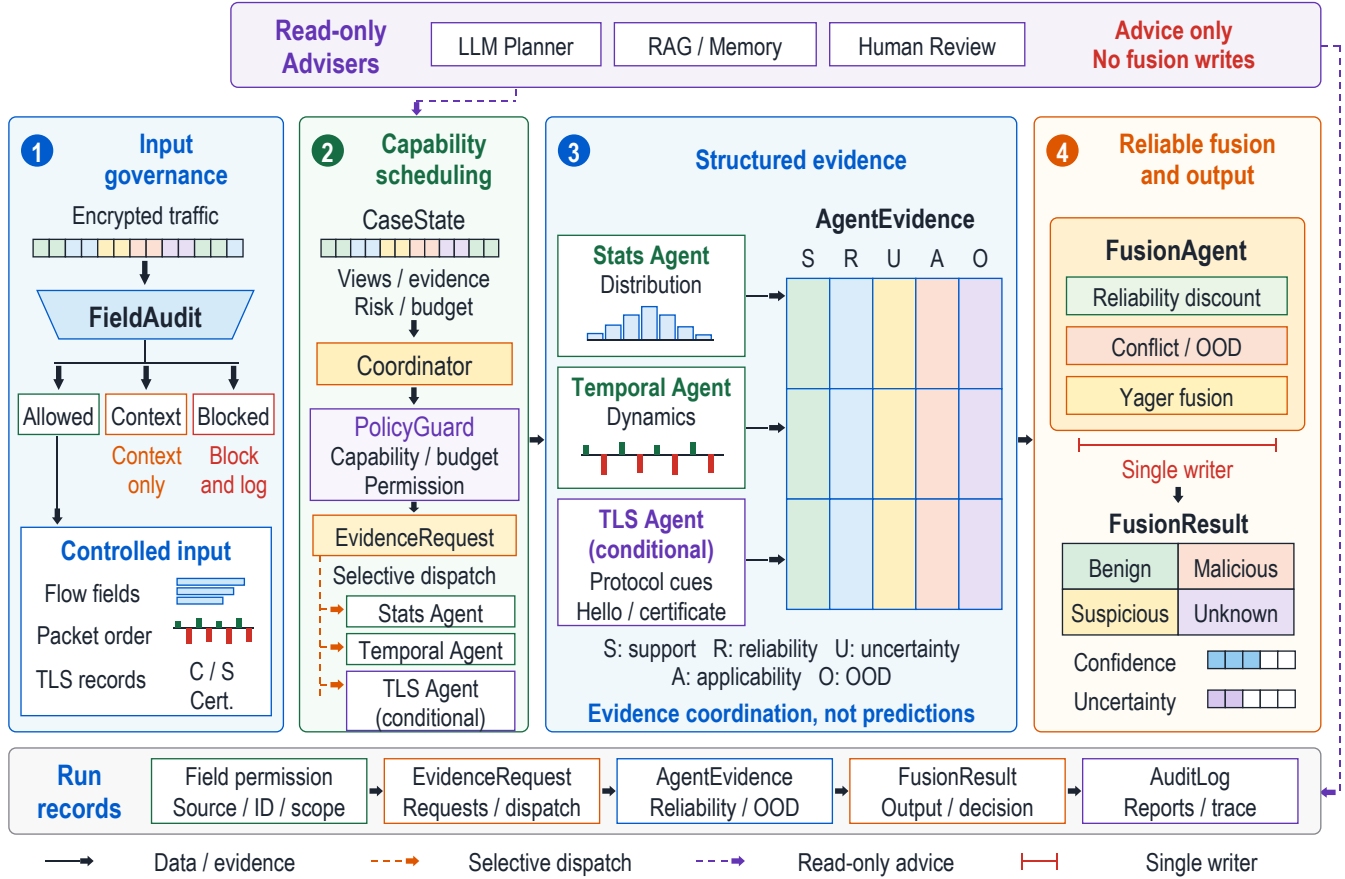


Fig. 1. Complete system context. Only Temporal/Stats enter the binary comparison, with both views available. TLS, advisers, Yager fusion and four-way outputs are background, not evaluated mechanisms. Raw timing covers the two-expert path, not this entire system.

are $\{0, 0.25, 0.5, 0.75, 1\}$, ties toward 0.5; Dev-expert ties retain Temporal.

USTC scores pool five outer confusion matrices. Against Temporal, C/D counts corrected/introduced errors; $C + D$ counts switches (no abstention). Net corrections differ from F1 change. Recall concerns malicious traffic, false-positive rate (FPR) benign traffic. Calls excludes parsing, features and arbitration.

Intervals use 1,000 class-stratified bootstraps of 20 family groups (seed 20260920), conditional on original HGB fits and the exposed population. They exclude activity/refitting uncertainty and do not apply to ExtraTrees or raw execution.

2.4. Candidate behavior and raw-input checks

For candidates Θ , final policy h_θ and selection samples S , post-hoc replay measures

$$m_\Theta(S) = \sum_{x \in S} \mathbf{1}[\{|h_\theta(x) : \theta \in \Theta|\} > 1], \quad (2)$$

$$K_\Theta(S) = |\{(h_\theta(x))_{x \in S} : \theta \in \Theta\}|. \quad (3)$$

Here $m/|S|$ is predictive ambiguity [9] over our full/recall-feasible grid; K counts prediction vectors, not independent samples. Let F be feasible candidates, $\hat{\theta}$ the choice, and $E \subseteq F$ those sharing its selection vector. $J = |E|$; U counts their distinct outer vectors. For outer errors L , the gap has within-/between-pattern terms:

$$L(\hat{\theta}) - \min_F L = [L(\hat{\theta}) - \min_E L] + [\min_E L - \min_F L]. \quad (4)$$

Both terms are nonnegative; this post-hoc identity uses outer labels, not a deployment rule or generalization bound.

With fits frozen, low/high arms exclude or prioritize rows varying across either arbiter's 16 candidates; random arms sample normally. Each draws 2,000 rows without replacement at fixed group, capture and class quotas. Arbiters share subsets from three dependent hash-ranked seeds. Recall-filter removal preserves ranking. Choices precede exposed-data replay; original intervals do not apply.

Three mutations change labels/identifiers, endpoints, or provenance/unknown fields; scoring truth and roles remain fixed. Timing stress doubles duration and prefix inter-arrival times without retraining.

Raw execution uses 24 captures, stream keys, 60-s idle splits and first-source direction. Eligibility requires TCP/UDP keys or IPv4 `ip.proto`; inherited labels do not certify attacks. Routing excludes evaluation families from fitting. Policies parse once, calling Stats on triggers (both for averages). Single-thread batches ≤ 128 use training-row warmup, i9-14900HX and 32 GiB RAM. Timing includes serialization/verification, excludes loading/network waiting; cache/load is uncontrolled.

3. CONTROLLED RESULTS

3.1. Corrections versus introduced errors

With HGB experts, Logistic-0.5 raises net corrections from Fixed's 25 to 69 with lower C, D (Table 1, Fig. 2). Its Macro-F1 difference against Fixed is $+0.110$ $[-0.037, +0.357]$ percentage points (pp; conditional 95% interval); HGB-0.5 differs by -0.143 $[-0.549, +0.114]$ pp.

Development selection reduces Logistic's corrected false alarms from 85 to 8 and added misses from 5 to 0. Relative to Fixed, Logistic-Dev16 differs by -0.073 $[-0.540, +0.315]$ pp; HGB-Dev16 differs by -0.055 $[-0.530, +0.341]$ pp.

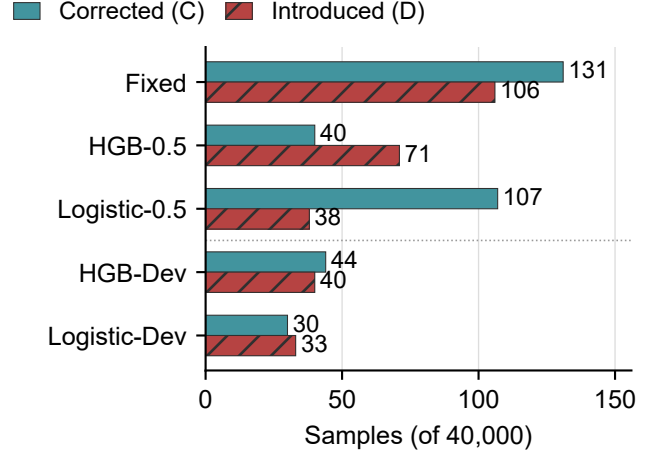


Fig. 2. Original HGB experts: absolute C, D (Dev = Dev16).

3.2. Candidate distinguishability and error transfer

Expert disagreement need not distinguish candidates (Table 3): fold 4's single disagreement gives $m = 0, K = 1$. All original selection disagreements are malicious. In fold 1, recall feasibility reduces Logistic's m/K from 6/9 to 4/4.

Fold 1 selects on MySQL/Neris and evaluates SMB/Weibo/Nsisay/Virut. Logistic's unique numerical optimum, $(0.25, 1.01)$, gains 0.075 pp on selection but worsens the outer error tradeoff versus 0.5 (Table 2); HGB improves on both roles. Conversely, fold 2's 16 selection-equivalent HGB candidates form six outer patterns.

Concentrated fitting gives Logistic fold 1 one feasible selection pattern but three outer patterns (-0.924 pp versus 0.5); HGB changes by zero there and -0.225 in fold 3. Over their respective predictions/feasible sets, full and concentrated Logistic fits give within/between gaps of 0+74 and 3+0.

Composition-matched coverage. Only fold 1 selects different low/high-coverage policies: $(0.25, 0.25)$ versus $(0.25, 1.01)$ for both arbiters. High coverage has more Logistic errors but fewer HGB errors (Table 2). All three dependent seeds agree; high equals random in outer predictions. Fold 3 reaches $m/K = 70/2$ without changing its choice; other folds retain 0/1.

Removing the recall filter changes none of 90 choices; the original filter excludes 0.5 in five cases. In these subsets, all candidates share benign predictions. At fixed FP, Macro-F1 increases with TP; the no-switch candidate matches Temporal, so every maximum-F1 candidate satisfies its recall.

Backend sensitivity. Stats ExtraTrees gives fold-1 selection gains of 0.150/0.075 pp (Logistic/HGB) but outer losses (Table 4): Logistic removes one miss and adds 80 false alarms; HGB adds 79 with unchanged misses. Eight other transfers are zero; no numeric optimum is unique. Logistic's feasible-candidate gap is zero; 0.5 is outside F (recall 0.993 versus Temporal's 0.995). Fold 3 diverges among ties without transfer loss.

Simple controls and sensitivity. Fixed-Dev16 minus Fixed gives -0.063 $[-0.537, +0.277]$ pp. Against Fixed-Dev16, Logistic-Dev16 gives -0.010 $[-0.050, +0.030]$ pp; HGB-Dev16 gives $+0.008$ $[-0.023, +0.045]$ pp. Each Global4 removes three false alarms from Dev16 with unchanged misses/corrections. Directional and equal-budget searches show no consistent advantage.

ExtraTrees lowers standalone Stats Macro-F1 but raises Fixed

Left minus right	Arbiter	Fold	Selection evidence	ΔF	ΔFN	$\Delta F1$
Dev16 – 0.5	Logistic	1	Unique numeric optimum ($J = 1$)	+77	–5	–0.899
High – low	Logistic	1	$m/K : 0/1 \rightarrow 6/9$	+78	–9	–0.862
High – low	HGB	1	$m/K : 0/1 \rightarrow 4/4$	+1	–31	+0.376
High – low	Both	3	$m/K : 0/1 \rightarrow 70/2$	0	0	0.000
No recall filter – original	Both	0–4	0/90 choices change	0	0	0.000

Table 2. HGB experts; stated contrasts use 8,000 outer samples/fold. F1: Macro-F1 pp; selection $n = 4,000$ in row 1, 2,000 otherwise; m/K : all 16 candidates. Coverage rows agree across three dependent seeds; fold 3 holds per arbiter; row 5 totals 90 paired conditions.

Fold	Fit	Dis.	Logistic						HGB					
			All m/K	Feas.	J/U	ΔF	C_{tr}/D_{tr}		All m/K	Feas.	J/U	ΔF	C_{tr}/D_{tr}	
0	1021	0	0/1	0/1	16/2	+0.000	0/0		0/1	0/1	16/1	+0.000	0/0	
1	6564	9	6/9	4/4	1/1	–0.899	5/77		4/4	4/3	3/1	+0.376	31/1	
2	2383	0	0/1	0/1	16/1	+0.000	0/0		0/1	0/1	16/6	+0.011	1/0	
3	3203	70	70/2	70/2	12/2	+0.000	0/0		70/2	70/2	12/6	+0.050	4/0	
4	1702	1	0/1	0/1	16/1	+0.000	0/0		0/1	0/1	16/4	+0.000	0/0	

Table 3. HGB replay. Fit: OOF disagreements; Dis.: triggered selection disagreements ($n = 4,000$). All/Feas.: full/feasible m/K ; J/U : equivalent candidates/outer patterns. ΔF : Dev16 minus arbiter-0.5 (Macro-F1 pp). C_{tr}/D_{tr} : errors corrected/introduced relative to arbiter-0.5.

Stats	Arbiter	Fold	J	U	Within	Between	ΔF
HGB	Logistic	1	1	1	0	74	–0.899
ExtraTrees	Logistic	1	3	1	0	0	–0.987
ExtraTrees	HGB	1	3	2	2	79	–0.987
ExtraTrees	Logistic	3	16	2	30	0	0.000
ExtraTrees	HGB	3	16	4	30	0	0.000

Table 4. Cases, not independent replications. Within/Between: gaps in (4); ΔF : Macro-F1 pp versus 0.5 (infeasible for ExtraTrees Logistic, fold 1).

Policy	Macro-F1			
	(%)	FP	FN	Wall (s)
Temporal	97.202	16033	1450	223.2
Tuned single	97.211	16047	1383	218.2
Equal average	96.355	19583	3191	262.0
Fixed	97.067	15222	3109	313.9
Logistic-0.5	97.362	15138	1350	291.3
HGB-0.5	96.914	16990	2295	411.8
Logistic-Dev16	97.130	16634	1297	388.3
HGB-Dev16	97.128	16772	1170	283.7

Table 5. Raw HGB execution: 625,523 segments (309,887 benign/315,636 malicious capture labels). Macro-F1 pools counts. Wall: one raw-to-output run, not online latency. Tuned single uses an earlier budget, not Fixed-Dev16.

above both learned 0.5 policies (Table 1).

Fitting budgets round up nonempty group–Temporal-decision–Stats-correctness cells. Source contrasts match counts, labels, directions, targets and inner models, concentrating each cell in its largest group or dispersing across groups, not randomized activities.

Larger fitting budgets do not monotonically reduce D . Dispersion gives HGB-0.5 seed C -0.063 pp, interval $[-0.141, -0.007]$. After selection, mean D falls from 62.00 to 37.67 and C from 46.00 to 43.33; Logistic is unchanged. HGB-Dev16 seed A gives $+0.095$ pp, interval $[-0.000276, +0.246]$. Intervals concern individual seeds, not their means.

Input checks and raw execution. All 120,000 mutations preserve features, probabilities, triggers, arbiter scores and classes. Timing stress changes 660 Temporal decisions (F1: 91.218%). Eight raw policies share 625,523 scored/9,107 unsupported IDs and class parity on 39,428 old segments. The 572 unsupported old segments (569 ARP, one ARP/Ethernet, two ICMPv6) are inherited-malicious and formerly used generic features. Logistic-0.5’s 92.915% on common segments versus 97.362% at natural frequencies reflects population change; it runs slower than Temporal (Table 5).

4. DISCUSSION AND CONCLUSION

Despite class balance, matched-quota subsets have no benign candidate variation; removing the recall filter leaves their choices un-

changed. Projection invariance leaves acquisition cues possible; timing stress does not identify shortcuts. Logistic-0.5’s Macro-F1 difference interval against Fixed spans zero.

Differently budgeted controls give tuned Temporal 92.875% and joint-view HGB 92.582%, outside Table 1. On exposed CIC-IDS2017 [13] Friday (2,000 benign/2,000 malicious), Temporal with first-cleaned-packet, 128-token ET-BERT [14] gives 57.779% versus Fixed’s 60.260% at the same trigger; selection has two malicious segments. This is neither pooled nor a matched-window test.

Separately, all five frozen USTC bundles and 11 policies were applied without refitting or reselection to 1,159,418 label-matched IoT-23 author flows (465,990 benign) from seven captures with known dependencies [15]. Across the 55 configurations, pooled Macro-F1 was 22.463–40.554% and FPR 92.811–100%. This exposed-data transfer changes both source and sample unit (author flows, not 60-s segments). High FPR limits deployment and does not identify the failure cause.

These cases separate selection information from error transfer: even a unique numerical optimum can worsen the error tradeoff, and higher coverage can move two arbiters in opposite directions.

5. REFERENCES

- [1] Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann, “Shortcut learning in deep neural networks,” *Nature Machine Intelligence*, vol. 2, pp. 665–673, 2020.
- [2] Minmin Chen, Zhixiang Xu, Kilian Weinberger, Olivier Chapelle, and Dor Kedem, “Classifier cascade for minimizing feature evaluation cost,” in *Proceedings of the Fifteenth International Conference on Artificial Intelligence and Statistics*. 2012, vol. 22 of *Proceedings of Machine Learning Research*, pp. 218–226, PMLR.
- [3] Sachini Piyoni Ekanayake and Daphney Stavroula Zois, “Instance-wise feature acquisition with classifier selection option for structured data instances,” in *2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025.
- [4] Rafael M. O. Cruz, Robert Sabourin, George D. C. Cavalcanti, and Tsang Ing Ren, “META-DES: A dynamic ensemble selection framework using meta-learning,” *Pattern Recognition*, vol. 48, no. 5, pp. 1925–1935, 2015.
- [5] Shilo Daum, Tal Shapira, Anat Bremler-Barr, and David Hay, “Non-uniformity is all you need: Efficient and timely encrypted traffic classification with ECHO,” arXiv:2406.01852v3, 2024, Version 3, July 9, 2024.
- [6] Heng Zhang, Ziqian Chen, Wei Xia, Gang Xiong, Gaopeng Gou, Zhen Li, Guangyan Huang, and Yunpeng Li, “ANASETC: Automatic neural architecture search for encrypted traffic classification,” in *2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025, pp. 1–5.
- [7] Gavin C. Cawley and Nicola L. C. Talbot, “On over-fitting in model selection and subsequent selection bias in performance evaluation,” *Journal of Machine Learning Research*, vol. 11, no. 70, pp. 2079–2107, 2010.
- [8] Alexander D’Amour, Katherine Heller, Dan Moldovan, et al., “Underspecification presents challenges for credibility in modern machine learning,” *Journal of Machine Learning Research*, vol. 23, pp. 1–61, 2022.
- [9] Charles Marx, Flavio Calmon, and Berk Ustun, “Predictive multiplicity in classification,” in *Proceedings of the 37th International Conference on Machine Learning*, 2020, vol. 119 of *Proceedings of Machine Learning Research*, pp. 6765–6774.
- [10] Haowen Liu and Lei Zhang, “Expert selection in traffic detection: Code and aggregate results,” <https://github.com/me-fake-you/traffic-expert-selection>, 2026, Research artifact accompanying an author-review manuscript.
- [11] USTC-TFC2016 contributors, “USTC-TFC2016 traffic dataset,” <https://github.com/davidyslu/USTC-TFC2016>, Dataset repository description, accessed September 20, 2026.
- [12] Pierre Geurts, Damien Ernst, and Louis Wehenkel, “Extremely randomized trees,” *Machine Learning*, vol. 63, pp. 3–42, 2006.
- [13] Canadian Institute for Cybersecurity, “Intrusion detection evaluation dataset (CIC-IDS2017),” <https://www.unb.ca/cic/datasets/ids-2017.html>, Dataset description, accessed September 20, 2026.
- [14] Xinjie Lin, Gang Xiong, Gaopeng Gou, Zhen Li, Junzheng Shi, and Jing Yu, “ET-BERT: A contextualized datagram representation with pre-training transformers for encrypted traffic classification,” arXiv:2202.06335v2, 2022, Accepted at The Web Conference 2022.
- [15] Sebastian Garcia, Agustin Parmisano, and Maria Jose Erquiaga, “IoT-23: A labeled dataset with malicious and benign IoT network traffic,” Zenodo, 2020, Version 1.0.0.